

Advanced Microeconomics

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Problem Set 5: Mathematical solutions

1) The utility function of an individual is given by $u(x_1, x_2) = x_1^{\frac{1}{3}} x_2^{\frac{2}{3}}$. The wealth of the individual and the prices of good 1 and 2 are greater than 0 ($w, p_1, p_2 > 0$).

a) Determine the Marshallian demand functions.

Solution:

$$x_1 = \frac{w}{3p_1}$$

$$x_2 = \frac{2w}{3p_2}$$

b) Determine the indirect utility function.

Solution:

$$v_{p_1, p_2, w} = \left(\frac{w}{3p_1} \right)^{\frac{1}{3}} \left(\frac{2w}{3p_2} \right)^{\frac{2}{3}}$$

c) Determine the expenditure function and the Hicksian demand functions.

Solution:

$$e_{p_1, p_2, u} = \frac{u}{\left(\frac{1}{3p_1} \right)^{\frac{1}{3}} \left(\frac{2}{3p_2} \right)^{\frac{2}{3}}}$$

$$h_1 = u \left(\frac{p_2}{2p_1} \right)^{\frac{2}{3}}; h_2 = u \left(\frac{2p_1}{p_2} \right)^{\frac{1}{3}}$$

2) Let the indirect utility function of an individual be $V(p_x, p_y, w) = \frac{w^2}{p_x p_y}$. Let income be $w > 0$ and prices for good x and good y, respectively, be $p_x, p_y > 0$.

a) Determine the individual's expenditure function.

Solution:

$$e(p_x, p_y, u) = \sqrt{p_x p_y u}$$

- b) Determine the Hicksian demands $h_x(p_x, p_y, U)$ and $h_y(p_x, p_y, U)$.

Solution:

$$h_x(p_x, p_y, u) = \frac{\sqrt{p_y u}}{2\sqrt{p_x}}$$

$$h_y(p_x, p_y, u) = \frac{\sqrt{p_x u}}{2\sqrt{p_y}}$$

- c) Determine the Marshallian demands $d_x(p_x, p_y, I)$ and $d_y(p_x, p_y, I)$ from the previous results.

Solution:

$$d_x(p_1, p_2, w) = \frac{w}{2p_x}$$

$$d_y(p_1, p_2, w) = \frac{w}{2p_y}$$

- d) Determine the underlying utility function of the individual.

Solution:

$$u(x, y) = 4xy$$